



Project initialization and planning phase

Date	2 July 2026
Team Id	XXXXXX
Project Name	Opti Crop : Smart Agricultural Production Optimization Engine
Maximum marks	5 marks

Project proposal report (proposed solution) report

The Smart Agricultural Production Optimization Engine is an AI-based crop recommendation system that helps farmers select the most suitable crop based on soil nutrients (Nitrogen, Phosphorus, Potassium), temperature, humidity, pH, and rainfall. The system analyses the input data using a trained Machine Learning model and provides accurate crop recommendations. It enables farmers to make informed decisions, improve crop yield, reduce resource wastage, and promote sustainable farming practices through a simple and user-friendly web application.

Project overview	
objective	The primary objective is to develop an intelligent crop recommendation system that predicts the most suitable crop using soil and climate data. The system enables farmers to make informed decisions, increase agricultural productivity, and promote sustainable farming practices.
scope	The project covers data collection, preprocessing, machine learning model development, prediction, and deployment of a web application. Users can input agricultural parameters and receive instant crop recommendations with a simple and user-friendly interface.

Problem statement	
Description	Farmers often face challenges in selecting the appropriate crop due to changing climatic conditions, lack of soil analysis, and insufficient agricultural guidance. Incorrect crop selection results in reduced yield, financial losses, and inefficient resource utilization.
Impact	Sub-optimal crop selection leads to lower agricultural productivity, increased production costs, excessive use of fertilizers and water, and reduced farmer income. These issues negatively affect sustainable agriculture and food security.

Proposed solution	
approach	Develop a Machine Learning-based web application that analyzes soil nutrients and environmental conditions to recommend the most suitable crop. The system uses a trained prediction model integrated with a Flask-based web application to deliver fast and accurate recommendations.
Key Features	- Machine learning – based crop recommendation system - Real time prediction using soil and climate parameters - Interactive flask web application with responsive user interface - Data pre-processing, visualization ,and model evaluation

Resource requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU specifications, number of cores	Intel Core i5 / AMD Ryzen 5 (or higher)
Memory	RAM specifications	8 GB RAM
Storage	Disk space for data, models, and project files	20 GB Free SSD/HDD
Software		
Frameworks	Python web framework	Flask
Libraries	Machine Learning and Data Processing Libraries	Scikit-learn, Pandas, NumPy, Joblib, Matplotlib



Development Environment	IDE	Visual Studio Code / PyCharm / Jupyter Notebook
Data		
Dataset	Source, size, and format	Crop Recommendation Dataset (CSV), ~2200 records